

Helium

3

Helium has only two electrons but this simplicity is deceptive. To treat systems with two particles requires new concepts that also apply to multi-particle systems in many branches of physics, and it is very worthwhile to study them carefully using the example of helium. There is truth in the saying that atomic physicists count ‘one, two, many’ and a detailed understanding of the two-electron system is sufficient for much of the atomic structure in this book.¹

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3.1 The ground state of helium

Two electrons in the Coulomb potential of a charge Ze , e.g. the nucleus of an atom, obey a Schrödinger equation of the form

$$\left\{ \frac{-\hbar^2}{2m} \nabla_1^2 + \frac{-\hbar^2}{2m} \nabla_2^2 + \frac{e^2}{4\pi\epsilon_0} \left(-\frac{Z}{r_1} - \frac{Z}{r_2} + \frac{1}{r_{12}} \right) \right\} \psi = E\psi. \quad (3.1)$$

Here $r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$ is the distance between electron 1 and electron 2 and the electrostatic repulsion of electrons is proportional to $1/r_{12}$. Neglecting this mutual repulsion for the time being, we can write the equation as

$$(H_1 + H_2) \psi = E^{(0)} \psi, \quad (3.2)$$

where

$$H_1 \equiv \frac{-\hbar^2}{2m} \nabla_1^2 - \frac{Z e^2}{4\pi\epsilon_0 r_1} \quad (3.3)$$

and H_2 is a similar expression for electron 2. Writing the atomic wavefunction as a product of the wavefunctions for each electron, $\psi = \psi(1)\psi(2)$, allows us to separate eqn 3.2 into two single-electron Schrödinger equations:

$$H_1 \psi(1) = E_1 \psi(1) \quad (3.4)$$

and a similar equation for $\psi(2)$ with energy E_2 . The solutions of these one-electron equations are hydrogenic wavefunctions with energies given by the Rydberg formula. Helium has $Z = 2$ and in its ground state both electrons have energy $E_1 = E_2 = -4hcR_\infty = -54.4 \text{ eV}$. Thus the total energy of the atom (neglecting repulsion) is

$$E^{(0)} = E_1 + E_2 = -109 \text{ eV}. \quad (3.5)$$

¹This book considers only those multi-electron systems with one, or two, valence electrons ‘outside’ a spherically-symmetric core of charge.

Now we need to calculate the perturbation produced by the electron–electron repulsion. The system has the spatial wavefunction

$$\psi_{1s^2} = R_{1s}^{Z=2}(r_1) R_{1s}^{Z=2}(r_2) \times \frac{1}{4\pi}, \quad (3.6)$$

² $1/\sqrt{4\pi}$ is the angular part of an s-electron wavefunction.

where radial wavefunctions are defined in Table 2.2.² The expectation value of the repulsion is (see Section 3.3)

$$\frac{e^2}{4\pi\epsilon_0} \int_0^\infty \int_0^\infty \psi_{1s^2}^* \frac{1}{r_{12}} \psi_{1s^2} r_1^2 dr_1 r_2^2 dr_2 = 34 \text{ eV}. \quad (3.7)$$

Adding this to the (zeroth-order) estimate $E^{(0)}$ gives an energy of $E(1s^2) = -109 + 34 = -75 \text{ eV}$. It takes an energy of 75 eV to remove both electrons from a helium atom leaving a bare helium nucleus He^{++} —the second ionization energy. To go from He^+ to He^{++} takes 54.4 eV, so this estimate suggests that the first ionization energy (required to remove one electron from He to create He^+) is $\text{IE}(\text{He}) \simeq 75 - 54 \simeq 21 \text{ eV}$. But the expectation value in eqn 3.7 is not small compared to the binding energy and therefore the perturbation has a significant effect on the wavefunctions. The necessary adjustment of the wavefunctions can be accounted for by the variational method.³ This method gives a value close to the measured ionization energy 24.6 eV. Helium has the highest first ionization energy of all elements because of its closed $n = 1$ shell. For a plot of the ionization energies of the elements see Grant and Phillips (2001, Chapter 11, Fig. 18).⁴

³This is a standard quantum mechanical technique whose mathematical details are given in quantum texts. The essential principle of this technique is to find an expression for the energy in terms of a parameter—an effective atomic number in the case of helium—and then minimise the energy with respect to this parameter, i.e. study the variation in the energy as a function of the chosen parameter.

⁴This is accessible at <http://www.oup.co.uk/best.textbooks/physics/ephys/illustrations/> along with other illustrations of elementary quantum ideas.

⁵It is often said that ‘one electron is in a spin-up state and the other is spin-down’; what this really means is defined in the discussion of spin for the excited states of helium.

According to the Pauli exclusion principle, two electrons cannot have the same set of quantum numbers. Therefore there must be some additional quantum number associated with the two 1s-electrons in the ground state of helium—this is their **spin** (introduced in Section 2.3.1). The observed filling-up of the atomic (sub-)shells in the periodic table implies that two spin states are associated with each set of spatial quantum numbers n, l, m_l .⁵ However, electrostatic energies do not depend on spin and we can find the spatial wavefunctions separately from the problem of finding the spin eigenfunctions.

3.2 Excited states of helium

To find the energy of the excited states we use the same procedure as for the ground state—at first we neglect the mutual repulsion term and separate eqn 3.1 into two one-electron equations that have solutions⁶

$$\begin{aligned} u_{1s}(1) &= R_{1s}(r_1) \times \frac{1}{\sqrt{4\pi}}, \\ u_{nl}(2) &= R_{nl}(r_2) Y_{l,m}(\theta_2, \phi_2) \end{aligned}$$

for the configuration $1s nl$. The spatial part of the atomic wavefunction is the product

$$\psi_{\text{space}} = u_{1s}(1) u_{nl}(2). \quad (3.8)$$

⁶The spatial wavefunction u contains both radial and angular parts but the energy does not depend on the magnetic quantum number, so we drop m as a subscript on u . The repulsion from a spherically-symmetric 1s wavefunction does not depend on the orientation of the other electron. To show this mathematically we could carry m through all the calculations and examine the resulting angular integrals, but this is cumbersome.

Another wavefunction has the same energy, namely

$$\psi_{\text{space}} = u_{1s}(2)u_{nl}(1). \quad (3.9)$$

These two states are related by a permutation of the labels on the electrons, $1 \leftrightarrow 2$; the energy cannot depend on the labeling of identical particles so there is **exchange degeneracy**. To consider the effect of the repulsive term on this pair of wavefunctions with the same energy (degenerate states) we need degenerate perturbation theory. There are two approaches. The look-before-you-leap approach is first to form eigenstates of the perturbation from linear combinations of the initial states.⁷ In this new basis the determination of the eigenenergies of the states is simple. It is instructive, however, simply to press ahead and go through the algebra once.⁸

⁷This is guided by looking for eigenstates of symmetry operators that commute with the Hamiltonian for the interaction, as in Section 4.5.

⁸In the light of this experience one can take the short cut in future.

We rewrite the Schrödinger equation (eqn 3.1) as

$$(H_0 + H')\psi = E\psi, \quad (3.10)$$

where $H_0 = H_1 + H_2$, and we consider the mutual repulsion of the electrons $H' = e^2/4\pi\epsilon_0 r_{12}$ as a perturbation. We also rewrite eqn 3.2 as

$$H_0\psi = E^{(0)}\psi, \quad (3.11)$$

where $E^{(0)} = E_1 + E_2$ is the unperturbed energy. Subtraction of eqn 3.11 from eqn 3.10 gives the energy change produced by the perturbation, $\Delta E = E - E^{(0)}$, as

$$H'\psi = \Delta E \psi. \quad (3.12)$$

A general expression for the wavefunction with energy $E^{(0)}$ is a linear combination of expressions 3.8 and 3.9, with arbitrary constants a and b ,

$$\psi = a u_{1s}(1)u_{nl}(2) + b u_{1s}(2)u_{nl}(1). \quad (3.13)$$

Substitution into eqn 3.12, multiplication by either $u_{1s}^*(1)u_{nl}^*(2)$ or $u_{1s}^*(2)u_{nl}^*(1)$, and then integration over the spatial coordinates for each electron (r_1, θ_1, ϕ_1 and r_2, θ_2, ϕ_2) gives two coupled equations that we write as

$$\begin{pmatrix} J & K \\ K & J \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \Delta E \begin{pmatrix} a \\ b \end{pmatrix}. \quad (3.14)$$

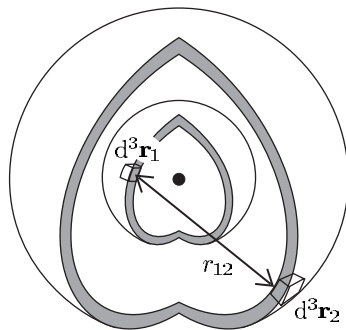
This is eqn 3.12 in matrix form. The direct integral is

$$\begin{aligned} J &= \frac{1}{4\pi\epsilon_0} \iint |u_{1s}(1)|^2 \frac{e^2}{r_{12}} |u_{nl}(2)|^2 \, \mathbf{dr}_1^3 \, \mathbf{dr}_2^3 \\ &= \frac{1}{4\pi\epsilon_0} \iint \frac{\rho_{1s}(r_1)\rho_{nl}(r_2)}{r_{12}} \, \mathbf{dr}_1^3 \, \mathbf{dr}_2^3, \end{aligned} \quad (3.15)$$

where $\rho_{1s}(1) = -e|u_{1s}(1)|^2$ is the charge density distribution for electron 1, and similarly for $\rho_{nl}(2)$. This direct integral represents the Coulomb repulsion of these charge clouds (Fig. 3.1). The exchange integral is

$$K = \frac{1}{4\pi\epsilon_0} \iint u_{1s}^*(1)u_{nl}^*(2) \frac{e^2}{r_{12}} u_{1s}(2)u_{nl}(1) \, \mathbf{dr}_1^3 \, \mathbf{dr}_2^3. \quad (3.16)$$

Fig. 3.1 The direct integral in a $1sns$ configuration of helium corresponds to the Coulomb repulsion between two spherically-symmetric charge clouds made up of shells of charge like those shown.



Unlike the direct integral, this does not have a simple classical interpretation in terms of charge (or probability) distributions—the exchange integral depends on interference of the amplitudes. The spherical symmetry of the $1s$ wavefunction makes the integrals straightforward to evaluate (Exercises 3.6 and 3.7).

The eigenvalues ΔE in eqn 3.14 are found from

$$\begin{vmatrix} J - \Delta E & K \\ K & J - \Delta E \end{vmatrix} = 0. \quad (3.17)$$

The roots of this determinantal equation are $\Delta E = J \pm K$. The direct integral shifts both levels together but the exchange integral leads to an energy splitting of $2K$ (see Fig. 3.2). Substitution back into eqn 3.14 gives the two eigenvectors in which $b = a$ and $b = -a$. These correspond to symmetric (S) and antisymmetric (A) wavefunctions:

$$\begin{aligned} \psi_{\text{space}}^S &= \frac{1}{\sqrt{2}} \{ u_{1s}(1)u_{nl}(2) + u_{1s}(2)u_{nl}(1) \}, \\ \psi_{\text{space}}^A &= \frac{1}{\sqrt{2}} \{ u_{1s}(1)u_{nl}(2) - u_{1s}(2)u_{nl}(1) \}. \end{aligned}$$

The wavefunction ψ_{space}^A has an eigenenergy of $E^{(0)} + J - K$, and this is lower than the energy $E^{(0)} + J + K$ for ψ_{space}^S . (For the $1snl$ configurations in helium K is positive.)⁹ This is often interpreted as the electrons ‘avoiding’ each other, i.e. $\psi_{\text{space}}^A = 0$ for $r_1 = r_2$, and for this wavefunction the probability of finding electron 1 close to electron 2 is small (see Exercise 3.3). This anticorrelation of the two electrons makes the expectation of the Coulomb repulsion between the electrons smaller than for ψ_{space}^S .

The occurrence of symmetric and antisymmetric wavefunctions has a classical analogue illustrated in Fig. 3.3. A system of two oscillators, with the same resonance frequency, that interact (e.g. they are joined together by a spring) has antisymmetric and symmetric normal modes as illustrated in Fig. 3.3(b) and (c). These modes and their frequencies are found in Appendix A as an example of the application of degenerate perturbation theory in *Newtonian mechanics*.¹⁰

The exchange integral decreases as n and l increase because of the reduced overlap between the wavefunctions of the excited electron and

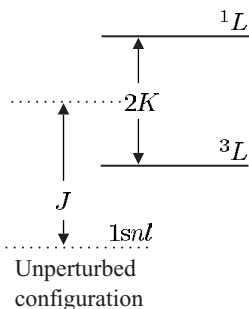


Fig. 3.2 The effect of the direct and exchange integrals on a $1snl$ configuration in helium. The singlet and triplet terms have an energy separation of twice the exchange integral ($2K$).

⁹It is easy to check which wavefunction corresponds to which eigenvalue by substitution into the original equation.

¹⁰Another example is the classical treatment of the normal Zeeman effect.

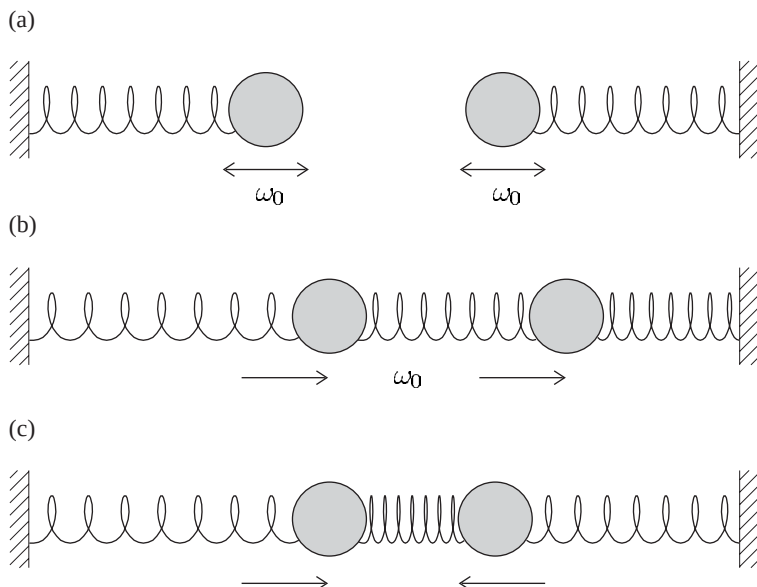


Fig. 3.3 An illustration of degenerate perturbation in a classical system. (a) Two harmonic oscillators with the same oscillation frequency ω_0 —each spring has a mass on one end and its other end is attached to a rigid support. An interaction, represented here by another spring that connects the masses, couples the motions of the two masses. The normal modes of the system are (b) an in-phase oscillation at ω_0 , in which the spring between the masses does not change length, and (c) an out-of-phase oscillation at a higher frequency. Appendix A gives the equations for this system of two masses and three springs, and also for the equivalent system of three masses joined by two springs that models a triatomic molecule, e.g. carbon dioxide.

the 1s-electron. These trends are an obvious consequence of the form of the wavefunctions: the excited electron's average orbit radius increases with energy and hence with n ; the variation with l arises because the effective potential from the angular momentum ('centrifugal' barrier) leads to the wavefunction of the excited electron being small at small r . However, in the treatment as described above, the direct integral does not tend to zero as n and l increase, as shown by the following physical argument. The excited electron 'sees' the nuclear charge of $+2e$ surrounded by the 1s electronic charge distribution, i.e. in the region far from the nucleus where nl -electron's wavefunction has a significant value it experiences a Coulomb potential of charge $+1e$. Thus the excited electron has an energy similar to that of an electron in the hydrogen atom, as shown in Fig. 3.4. But we have started with the assumption that both the 1s- and nl -electrons have an energy given by the Rydberg formula for $Z = 2$. The direct integral J equals the difference between these energies.¹¹ This work was an early triumph for wave mechanics since previously it had not been possible to calculate the structure of helium.¹²

In this section we found the wavefunctions and energy levels in helium by direct calculation but looking back we can see how to anticipate the answer by making use of symmetry arguments. The Hamiltonian for the electrostatic repulsion, proportional to $1/r_{12} \equiv 1/|\mathbf{r}_1 - \mathbf{r}_2|$, commutes with the operator that interchanges the particle labels 1 and 2, i.e. the swap operation $1 \leftrightarrow 2$. (Although we shall not give this operator a symbol it is obvious that it leaves the value of $1/r_{12}$ unchanged.) Commuting operators have simultaneous eigenfunctions. This prompts us

¹¹This can also be seen from eqn 3.15. The integration over r_1 , θ_1 and ϕ_1 leads to a repulsive Coulomb potential $\sim e/4\pi\epsilon_0 r_2$ that cancels part of the attractive potential of the nucleus, when r_2 is greater than the values of r_1 where ψ_1 is appreciable.

¹²For hydrogen, the solution of Schrödinger's equation reproduced the energy levels calculated by the Bohr-Sommerfeld theory. However, wave mechanics does give more information about hydrogen than the old quantum theory, e.g. it allows the detailed calculation of transition rates.

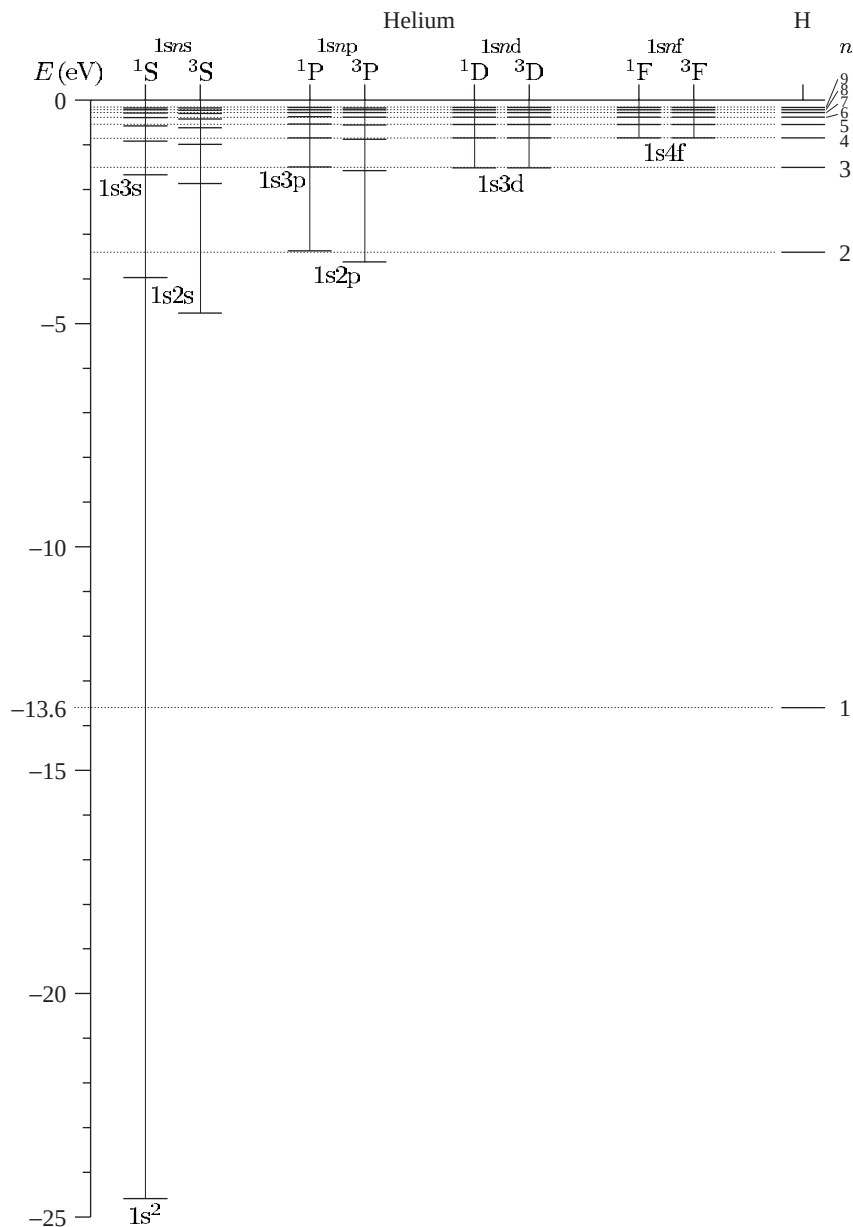


Fig. 3.4 The energy levels of the helium atom with those of hydrogen for comparison. The $1s^2$ ground configuration is tightly bound. For the excited configurations of helium the 1s-electron screens the outer electron from the nuclear charge so that the $1snl$ configurations in helium have similar energy to the shell with principal quantum number n in hydrogen. The hydrogenic levels are indicated on the right. The interval between the 1L and 3L terms (equal to twice the exchange integral) is clear for the $1s2s$, $1s2p$, $1s3s$, $1s3p$ and $1s4s$ configurations but it is smaller for higher n and l .

to construct the symmetrised wavefunctions $\psi_{\text{space}}^{\text{A}}$ and $\psi_{\text{space}}^{\text{S}}$.¹³ In this basis of eigenstates it is simple to calculate the effect of the electrostatic repulsion.

3.2.1 Spin eigenstates

The electrostatic repulsion between the two electrons leads to the wavefunctions $\psi_{\text{space}}^{\text{S}}$ and $\psi_{\text{space}}^{\text{A}}$ in the excited states of the helium atom. The ground state is a special case where both electrons have the same spatial wavefunction, so only a symmetric solution exists. We did not consider spin since electrostatic interactions depend on the charge of the particle, not their spin. Neither H_0 nor H' contains any reference to the spin of the electrons. Spin does, however, have a profound effect on atomic wavefunctions. This arises from the deep connection between spin and the *symmetry* of the wavefunction of indistinguishable particles.¹⁴ Note that here we are considering the total wavefunction in the systems that includes both the spatial part (found in the previous section) and the spin. Fermions have wavefunctions that are antisymmetric with respect to particle-label interchange, and bosons have symmetric ones. As a consequence of this symmetry property, fermions and bosons fill up the levels of a system in different ways, i.e. they obey different quantum statistics.

Electrons are fermions so atoms have total wavefunctions that are antisymmetric with respect to permutation of the electron labels. This requires $\psi_{\text{space}}^{\text{S}}$ to associate with an antisymmetric spin function $\psi_{\text{spin}}^{\text{A}}$, and the other way round:

$$\psi = \psi_{\text{space}}^{\text{S}} \psi_{\text{spin}}^{\text{A}} \quad \text{or} \quad \psi_{\text{space}}^{\text{A}} \psi_{\text{spin}}^{\text{S}}. \quad (3.18)$$

These antisymmetrised wavefunctions that we have constructed fulfil the requirement of having particular symmetry with respect to the interchange of indistinguishable particles. Now we shall find the spin eigenfunctions explicitly. We use the shorthand notation where $\uparrow$ and $\downarrow$ represent $m_s = 1/2$ and $-1/2$, respectively. Two electrons have four possible combinations: the three symmetric functions,

$$\begin{aligned} \psi_{\text{spin}}^{\text{S}} &= |\uparrow\uparrow\rangle \\ &= \frac{1}{\sqrt{2}} \{ |\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle \} \\ &= |\downarrow\downarrow\rangle, \end{aligned} \quad (3.19)$$

corresponding to $S = 1$ and $M_S = +1, 0, -1$; and an antisymmetric function

$$\psi_{\text{spin}}^{\text{A}} = \frac{1}{\sqrt{2}} \{ |\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \}, \quad (3.20)$$

corresponding to $S = 0$ (with $M_S = 0$).¹⁵ Spectroscopists label the eigenstates of the electrostatic interactions with the symbol ^{2S+1}L , where S is the total spin and L is the total orbital angular momentum quantum number. The $1snl$ configurations in helium $L = l$, so the allowed terms

¹³For two electrons, swapping the particle labels twice brings us back to where we started, so $\psi(1,2) = \pm\psi(2,1)$. Therefore the two possible eigenvalues are 1 for $\psi_{\text{space}}^{\text{S}}$ and -1 for $\psi_{\text{space}}^{\text{A}}$.

¹⁴Indistinguishable means that the particles are identical and have the freedom to exchange positions, e.g. atoms in a gas which obey Fermi–Dirac or Bose–Einstein statistics depending on their spin. In contrast, atoms in a solid can be treated as distinguishable, even if they are identical, because they have fixed positions—we could label the atoms 1, 2, etc. and still know which is which at some later time.

¹⁵These statements about the result of adding two $s = 1/2$ angular momenta can be proved by formal angular momentum theory. Simplified treatments describe $S = 0$ as having one electron with ‘spin-up’ and the other with ‘spin-down’; but both $M_S = 0$ states are linear combinations of the states $|m_{s1} = +1/2, m_{s2} = -1/2\rangle$ and $|m_{s1} = -1/2, m_{s2} = +1/2\rangle$.

¹⁶The letter ‘S’ appears over-used in this established notation but no ambiguity arises in practice. The symbol S for the total spin is italic because this is a variable, whereas the symbols S for $L = 0$ and s for $l = 0$ are not italic.

are 1L and 3L , e.g. the $1s2s$ configuration in helium gives rise to the terms 1S and 3S , where S represents $L = 0$.¹⁶

In summary, we have calculated the structure of helium in two distinct stages.

- (1) **Energies** Degenerate perturbation theory gives the space wavefunctions ψ_{space}^S and ψ_{space}^A with energies split by twice the exchange integral. In helium the degeneracy arises because the two electrons are identical particles so there is exchange degeneracy, but the treatment is similar for systems where a degeneracy arises by accident.
- (2) **Spin** We determined the spin associated with each energy level by constructing symmetrised wavefunctions. The product of the spatial functions and the spin eigenstates gives the total atomic wavefunction that must be antisymmetric with respect to particle-label interchange.

Exchange degeneracy, exchange integrals, degenerate perturbation theory and symmetrised wavefunctions all occur in helium and their interrelationship is not straightforward so that misconceptions abound. A common misinterpretation is to infer that because levels with different total spin, $S = 0$ and 1 , have different energies then there is a spin-dependent interaction—this is *not* correct, but sometimes in condensed matter physics it is useful to pretend that it is! (See Blundell 2001.) The interactions that determine the gross structure of helium are entirely *electrostatic* and depend only on the charge and position of the particles. Also, degenerate perturbation theory is sometimes regarded as a mysterious quantum phenomenon. Appendix A gives further discussion and shows that symmetric and antisymmetric normal modes occur when two classical systems, with similar energy, interact, e.g. two coupled oscillators.

3.2.2 Transitions in helium

To determine which transitions are allowed between the energy levels of helium we need a selection rule for spin: the total spin quantum number does not change in electric dipole transitions. In the matrix element $\langle \psi_{\text{final}} | r | \psi_{\text{initial}} \rangle$ the operator r does not act on spin; therefore, if the ψ_{final} and ψ_{initial} do not have the same value of S , then their spin functions are orthogonal and the matrix element equals zero.¹⁷ This selection rule gives the transitions shown in Fig. 3.5.

¹⁷This anticipates a more general discussion of this and other selection rules for the LS -coupling scheme in a later chapter.

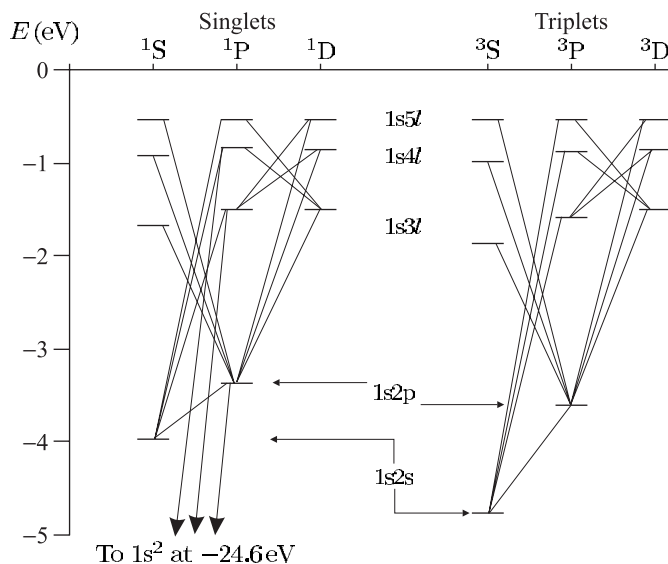


Fig. 3.5 The allowed transitions between the terms of helium are governed by the selection rule $\Delta S = 0$ in addition to the rule $\Delta l = \pm 1$ found previously. Since there are no transitions between singlets and triplets it is convenient to draw them as two separate systems. Notice that in the radiative decay of helium atoms excited to high-lying levels there are bottlenecks in the metastable $1s2s$ 1S and $1s2s$ 3S terms.

3.3 Evaluation of the integrals in helium

In this section we shall calculate the direct and exchange integrals to make quantitative predictions for some of the energy levels in the helium atom, based on the theory described in the previous sections. This provides an example of the use of atomic wavefunctions to carry out a calculation where there are no corresponding classical orbits and gives an indication of the complexities that arise in systems with more than one electron. The evaluation of the integrals requires care and some further details are given in Appendix B. The important point to be learnt from this section, however, is not the mathematical techniques but rather to see that the integrals arise from the Coulomb interaction between electrons treated by straightforward quantum mechanics.

3.3.1 Ground state

To calculate the energy of the $1s^2$ configuration we need to find the expectation value of $e^2/4\pi\epsilon_0 r_{12}$ in eqn 3.1—this calculation is the same as the evaluation of the mutual repulsion between two charge distributions in classical electrostatics, as in eqn 3.15 with $\rho_{1s}(r_1)$ and $\rho_{nl}(r_2) = \rho_{1s}(r_2)$. The integral can be considered in different ways. We could calculate the energy of the charge distribution of electron 1 in the potential created by electron 2, or the other way around. This section does neither; it uses a method that treats each electron symmetrically (as in Appendix B), but of course each approach gives the same numerical result. Electron 1 produces an electrostatic potential at radial distance r_2 given by

$$V_{12}(r_2) = \int_0^{r_2} \frac{1}{4\pi\epsilon_0 r_{12}} \rho(r_1) d^3\mathbf{r}_1. \quad (3.21)$$

The spherical symmetry of s-electrons means that the charge in the region $r_1 < r_2$ acts like a point charge at the origin, so that

$$V_{12}(r_2) = \frac{Q(r_2)}{4\pi\epsilon_0 r_2},$$

where $Q(r_2)$ is the charge within a radius of r_2 from the origin, which is given by¹⁸

$$Q(r_2) = \int_0^{r_2} \rho(r_1) 4\pi r_1^2 dr_1. \quad (3.22)$$

The electrostatic energy that arises from the repulsion equals

$$E_{12} = \int_0^\infty V_{12}(r_2) \rho(r_2) 4\pi r_2^2 dr_2. \quad (3.23)$$

For the $1s^2$ configuration there is an exactly equal contribution to the energy from $V_{21}(r_1)$, the (partial) potential at r_1 produced by electron 2. Thus the total energy of the repulsion between the electrons is twice that in eqn 3.23.¹⁹ Using the radial wavefunction for a $1s$ -electron, we find

$$\begin{aligned} J_{1s^2} &= 2 \times \frac{e^2}{4\pi\epsilon_0} \int_0^\infty \left\{ \int_0^{r_2} \frac{1}{r_1} 4Z^3 e^{-(Z/a_0)2r_1} r_1^2 dr_1 \right\} 4Z^3 e^{-(Z/a_0)2r_2} r_2^2 dr_2 \\ &= \frac{e^2/4\pi\epsilon_0}{2a_0} \frac{5}{4} Z = (13.6 \text{ eV}) \times \frac{5}{4} Z. \end{aligned} \quad (3.24)$$

For helium this gives $J_{1s^2}^{Z=2} = 34 \text{ eV}$.

3.3.2 Excited states: the direct integral

A $1snl$ configuration of helium has an energy close to that of an nl -electron in hydrogen, e.g. in the $1s2p$ configuration the $2p$ -electron has a similar binding energy to the $n = 2$ shell of hydrogen. The obvious explanation, in Bohr's model, is that the $2p$ -electron lies outside the $1s$ -orbit so that the inner electron screens the outer one from the full nuclear charge. Applying an analogous argument to the quantum treatment of helium leads to the Hamiltonian $H = H_{0a} + H'_a$, where²⁰

$$H_{0a} = -\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - \frac{e^2}{4\pi\epsilon_0} \left(\frac{2}{r_1} + \frac{1}{r_2} \right) \quad (3.25)$$

and

$$H'_a = \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r_{12}} - \frac{1}{r_2} \right). \quad (3.26)$$

In the expression for H_{0a} , electron 2 experiences the Coulomb attraction of a charge $+1e$. In H'_a the subtraction of $e^2/4\pi\epsilon_0 r_2$ from the mutual repulsion means that the perturbation tends to zero at a large distance from the nucleus (which is intuitively reasonable). This decomposition differs from that in Section 3.1. The different treatment of the two electrons makes the perturbation theory a little tricky, but Heisenberg

¹⁸Here $Q(\infty) = -e$.

¹⁹As is usual in calculations of the interaction between electric charge distributions, one must be careful to avoid double counting. This method of calculation avoids this pitfall, as shown by the general argument in Appendix B. An alternative method is used in Woodgate (1980), Problem 5.5.

²⁰The effect of the repulsion proportional to $1/r_{12}$ can be considered in terms of potentials like that in eqn 3.21 (and Appendix B). The potential at the position of the outer electron r_2 arising from the charge distribution of electron 1 accounts for a large portion of the total repulsion: $V_{12}(r_2) \simeq e^2/4\pi\epsilon_0 r_2$ in the region where $\rho_{nl}(r_2)$ has an appreciable value. Hence it makes sense to include $e^2/4\pi\epsilon_0 r_2$ in the zeroth-order Hamiltonian H_{0a} and treat the (small) part left over as a perturbation H'_a .

did the calculation as described in Bethe and Salpeter (1957) or Bethe and Salpeter (1977); he found the direct integral

$$J_{1snl} = \frac{e^2}{4\pi\epsilon_0} \iint \left(\frac{1}{r_{12}} - \frac{1}{r_2} \right) |u_{1s}(1)|^2 |u_{nlm}(2)|^2 d^3\mathbf{r}_1 d^3\mathbf{r}_2. \quad (3.27)$$

This must be evaluated with the appropriate wavefunctions, i.e. $u_{nlm}^{Z=1}$ rather than $u_{nlm}^{Z=2}$, and $u_{1s}^{Z=2}$ as before.²¹ For the excited electron $u_{nlm} = R_{nl}(r)Y_{lm}(\theta, \phi)$, where $R_{nl}(r)$ is the radial function for $Z = 1$. We write the direct integral as

$$J_{1snl} = \frac{e^2}{4\pi\epsilon_0} \int_0^\infty \int_0^\infty J(r_1, r_2) R_{10}^2(r_1) R_{nl}^2(r_2) r_1^2 dr_1 r_2^2 dr_2, \quad (3.28)$$

where the angular parts are contained in the function²²

$$J(r_1, r_2) = \int_0^{2\pi} \int_0^\pi \int_0^{2\pi} \int_0^\pi \left(\frac{1}{r_{12}} - \frac{1}{r_2} \right) \frac{1}{4\pi} |Y_{lm}(\theta_2, \phi_2)|^2 \times \sin \theta_1 d\theta_1 d\phi_1 \sin \theta_2 d\theta_2 d\phi_2. \quad (3.29)$$

The calculation of this integral requires the expansion of $1/r_{12}$ in terms of spherical harmonics:²³

$$\frac{1}{r_{12}} = \frac{1}{r_2} \sum_{k=0}^\infty \left(\frac{r_1}{r_2} \right)^k \frac{4\pi}{2k+1} \sum_{q=-k}^k Y_{k,q}^*(\theta_1, \phi_1) Y_{k,q}(\theta_2, \phi_2) \quad (3.30)$$

for $r_2 > r_1$ (and $r_1 \leftrightarrow r_2$ when $r_1 > r_2$). Only the term for $k = 0$ survives in the integration over angles in eqn 3.29 to give²⁴

$$J(r_1, r_2) = \begin{cases} 0 & \text{for } r_1 < r_2, \\ 1/r_1 - 1/r_2 & \text{for } r_1 > r_2. \end{cases}$$

When $r_1 < r_2$ the original screening argument applies and eqn 3.25 gives a good description. When $r_1 > r_2$ the appropriate potential is proportional to $-2/r_2 - 1/r_1$ and $J(r_1, r_2)$ accounts for the difference between this and $-2/r_1 - 1/r_2$ used in H_{0a} . Thus we find

$$J_{1snl} = \frac{e^2}{4\pi\epsilon_0} \int_0^\infty \left\{ \int_{r_2}^\infty \left(\frac{1}{r_1} - \frac{1}{r_2} \right) R_{10}^2(r_1) r_1^2 dr_1 \right\} R_{nl}^2(r_2) r_2^2 dr_2. \quad (3.31)$$

Evaluation of this integral for the 1s2p configuration (in Exercise 3.6) gives $J_{1s2p} = -2.8 \times 10^{-2}$ eV—three orders of magnitude smaller than $J_{1s^2}^{Z=2}$ in eqn 3.7 (evaluated from eqn 3.24). The unperturbed wavefunction for $Z = 1$ has energy equal to that of the corresponding level in hydrogen and the small negative direct integral accounts for the incompleteness of the screening of the nl -electron by the inner electron.

3.3.3 Excited states: the exchange integral

The exchange integral has the same form as eqn 3.16 but with $u_{nlm}^{Z=1}$ rather than $u_{nlm}^{Z=2}$ (and $u_{1s}^{Z=2}$ as before). Within the spatial wavefunction

²¹We have not derived this integral rigorously but it has an intuitively reasonable form.

²² $Y_{00}(\theta_1, \phi_1) = 1/\sqrt{4\pi}$.

²³ $Y_{k,q}^*(\theta_1, \phi_1) = (-1)^q Y_{k,-q}(\theta_1, \phi_1)$.

²⁴When $k \neq 0$ the integral of the function $Y_{k,q}^*(\theta_1, \phi_1)$ over θ_1 and ϕ_1 equals zero.

$u_{nlm} = R_{nl}(r)Y_{lm}(\theta, \phi)$ only the radial part depends on Z . We write the exchange integral as (cf. eqn 3.28)

$$K_{1snl} = \frac{e^2}{4\pi\epsilon_0} \iint K(r_1, r_2) R_{1s}(r_1) R_{nl}(r_1) R_{1s}(r_2) R_{nl}(r_2) r_1^2 dr_1 r_2^2 dr_2. \quad (3.32)$$

The function $K(r_1, r_2)$ containing the angular integrals is (cf. eqn 3.29)

$$K(r_1, r_2) = \iiint \frac{1}{r_{12}} Y_{lm}^*(\theta_1, \phi_1) \frac{1}{4\pi} Y_{lm}(\theta_2, \phi_2) \times \sin \theta_1 d\theta_1 d\phi_1 \sin \theta_2 d\theta_2 d\phi_2. \quad (3.33)$$

For the $1snp$ configuration only the second term of the expansion in eqn 3.30, with $k = 1$, survives in the integration because of the orthogonality of the spherical harmonic functions (see Exercise 3.7), to give

$$K(r_1, r_2) = \begin{cases} r_1/3r_2^2 & \text{for } r_1 < r_2, \\ r_2/3r_1^2 & \text{for } r_2 < r_1. \end{cases} \quad (3.34)$$

Carrying out the integration over the radial wavefunctions in eqn 3.32 for the $1s2p$ configuration gives the splitting between 3P and 1P as $2K_{1s2p} \simeq 0.21 \text{ eV}$ (close to the measured value of 0.25 eV).

The assumption that the excited electron lies outside the $1s$ wavefunction does not work so well for $1sns$ configurations since $\psi_{ns}(0)$ has a finite value and the above method of calculating J and K is less accurate.²⁵ The $1s2s$ configuration of helium has a singlet–triplet separation of $E(^1S) - E(^3S) = 2K_{1s2s} \simeq 0.80 \text{ eV}$ and the direct integral is also larger than that for $1s2p$ —these trends are evident in Fig. 3.4 (see also Exercise 3.7).²⁶

In some respects, helium is a more typical atom than hydrogen. The Schrödinger and Dirac equations can be solved exactly for the one-electron system, but not for helium or other atoms with more electrons. Thus in a careful study of helium we encounter the approximations needed to treat multi-electron atoms, and this is very important for understanding atomic structure in general. Helium also gives a good example of the influence of identical particles on the occupation of the states in quantum systems. The energy levels of the helium atom (and the existence of exchange integrals) do not depend on the fact that the two electrons are identical, as demonstrated in Exercises 3.3 and 3.4; however, this is a common point of confusion. The books recommended for further reading give clear and accurate descriptions of helium that reward careful study.

Further reading

The recommended books are divided into two categories corresponding to the two main themes in this chapter: (a) a description of how to calculate the electrostatic energy in an atom with more than one electron, which introduces principles that can be used in atoms with more

²⁵At small r the wavefunction of an ns -electron deviates significantly from $u_{ns}^{Z=1}$; for this reason $1s2p$ was chosen as an example above.

²⁶The overlap of the $1s$ and nl wavefunctions becomes smaller as n and l increase. In Heisenberg's treatment where screening is taken into account, the direct integral gives the deviation from the hydrogenic levels (which could be characterised by a quantum defect as in the alkalis, see Chapter 4). For electrons with $l \neq 0$ the term $\hbar^2 l(l+1)/2mr^2$ in the Schrödinger equation causes the electron's wavefunction to lie almost entirely outside the region where $u_{1s}^{Z=2} = R_{1s}(r)/\sqrt{4\pi}$ has a significant value.

electrons; and (b) a discussion of the influence of identical particles on the statistics of a quantum system that is important throughout physics. The influence of identical particles on the occupation of the quantum levels of a system with many particles, i.e. Bose–Einstein and Fermi–Dirac statistics, is discussed in statistical mechanics texts. Clear descriptions of helium may be found in the following textbooks: Cohen-Tannoudji *et al.* (1977), Woodgate (1980) and Mandl (1992). The calculation of the direct and exchange integrals in Section 3.3 is based on the definitive work by Bethe and Salpeter (1957), or see Bethe and Jackiw (1986).

A very instructive comparison can be made between the properties of the two electrons in helium and the nuclear spin statistics of homonuclear diatomic molecules²⁷ described in Atkins (1983, 1994).²⁸ There are diatomic molecules with nuclei that are identical bosons, identical fermions and cases of two similar but not identical particles, and their study gives a wider perspective than the study of helium alone. The nuclei of the two atoms in a hydrogen molecule are protons which are fermions (like the two electrons in helium).²⁹ For reasons explained in the above references, we can consider only those parts of the molecular wavefunction that describe the rotation ψ_{rot} and the nuclear spin states ψ_I —these are spatial and spin wavefunctions, respectively. For H_2 the wavefunction must have overall antisymmetry with respect to an interchange of particle labels since the nuclei are protons, each with a spin of $1/2$. This requires that a rotational must pair with a spin function of the opposite symmetry:

$$\psi_{\text{molecule}} = \psi_{\text{rot}}^S \psi_I^A \quad \text{or} \quad \psi_{\text{rot}}^A \psi_I^S. \quad (3.35)$$

This is analogous to eqn 3.18 for helium; as described in Section 3.2.1, the two spin- $1/2$ nuclei in a hydrogen molecule give a total (nuclear) spin of 0 and 1, with one state and three states, respectively. The 1 to 3 ratio of the number of nuclear spins associated with the energy levels for ψ_{rot}^S and ψ_{rot}^A , respectively, influences the populations of these rotational energy levels in a way that is directly observed in molecular spectra (the intensity of the lines in spectra depends on the population of the initial level). The molecule HD made from hydrogen and deuterium does not have identical nuclei so there is no overall symmetry requirement, but it has similar energy levels to those of H_2 apart from the mass dependence. This gives a real physical example where the statistics depends on whether the particles are identical or not, but the energy of the system does not. Exercise 3.4 discusses an artificial example: a helium-like system that has the same energy levels as a helium atom and hence the same direct and *exchange* integrals, even though the constituent particles are not identical.

²⁷Molecules made up of two atoms with identical nuclei.

²⁸These books also summarise the helium atom and the quantum mechanics of these molecular systems is very closely related to atomic physics.

²⁹The wavefunction of the hydrogen molecule has exchange symmetry—crudely speaking, the molecule looks the same when rotated through 180° .

Exercises

More advanced problems are indicated by a *.

(3.1) Estimate of the binding energy of helium

- Write down the Schrödinger equation for the helium atom and state the physical significance of each of the terms.
- Estimate the equilibrium energy of an electron bound to a charge $+Ze$ by minimising

$$E(r) = \frac{\hbar^2}{2mr^2} - \frac{Ze^2}{4\pi\epsilon_0 r}.$$

- Calculate the repulsive energy between the two electrons in helium assuming that $r_{12} \sim r$. Hence estimate the ionization energy of helium.
- Estimate the energy required to remove a further electron from the helium-like ion Si^{12+} , taking into account the scaling with Z of the energy levels and the expectation value for the electrostatic repulsion. The experimental value is 2400 eV. Compare the accuracy of your estimates for Si^{12+} and helium. (IE(He) = 24.6 eV.)

(3.2) Direct and exchange integrals for an arbitrary system

- Verify that for

$$\begin{aligned} \psi^A(r_1, r_2) \\ = \frac{1}{\sqrt{2}} \{ u_\alpha(r_1)u_\beta(r_2) - u_\alpha(r_2)u_\beta(r_1) \} \end{aligned}$$

and $H' = e^2/4\pi\epsilon_0 r_{12}$ the expectation value $\langle \psi^A | H' | \psi^A \rangle$ has the form $J - K$ and give the expressions for J and K .

- Write down the wavefunction ψ^S that is orthogonal to ψ^A .
- Verify that $\langle \psi^A | H' | \psi^S \rangle = 0$ so that H' is diagonal in this basis.

(3.3) Exchange integrals for a delta-function interaction

A particle in a square-well potential, with $V(x) = 0$ for $0 < x < \ell$ and $V(x) = \infty$ elsewhere, has normalised eigenfunctions $u_0(x) = \sqrt{2/\ell} \sin(\pi x/\ell)$ and $u_1(x) = \sqrt{2/\ell} \sin(2\pi x/\ell)$.

- What are the eigenenergies E_0 and E_1 of these two wavefunctions for a particle of mass m ?

- Two particles of the same mass m are both in the ground state so that the energy of the whole system is $2E_0$. Calculate the perturbation produced by a point-like interaction described by the potential $a \delta(x_1 - x_2)$, with a constant.
- Show that, when the two interacting particles occupy the ground and first excited states, the direct and exchange integrals are equal. Also show that the delta-function interaction produces no energy shift for the antisymmetric spatial wavefunction and explain this in terms of correlation of the particles. Calculate the energy of the other level of the perturbed system.
- For the two energy levels found in part (c), sketch the spatial wavefunction as a function of the coordinates of the two particles x_1 and x_2 . The particles move in one dimension but the two-particle wavefunction exists in a two-dimensional Hilbert space—draw either a contour plot in the $x_1 x_2$ -plane or attempt a three-dimensional sketch (by hand or computer).
- The two particles are identical and have spin $1/2$. What is the total spin quantum number S associated with each of the energy levels found in part (c)?

- *(f) Discuss qualitatively the energy levels of this system for two particles that have slightly different masses $m_1 \neq m_2$, so that they are distinguishable? [Hint. The spin has not been given because it is not important for non-identical particles.]

Comment. The antisymmetric spatial wavefunction in part (c) clearly has different properties from a straightforward product $u_0 u_1$. The exchange integral is a manifestation of the entanglement of the multiple-particle system.

(3.4) A helium-like system with non-identical particles

Imagine that there exists an exotic particle with the same mass and charge as the electron but spin $3/2$ (so it is not identical to the electron). This particle and an electron form a bound system with a helium nucleus. Compare the energy levels of this system with those of the helium atom. Describe the energy levels of a system with two of the ex-

otic particles bound to a helium nucleus (and no electrons). [*Hint.* It is not necessary to specify the values of total spin associated with the levels.]

(3.5) *The integrals in helium*

- Show that the integral in eqn 3.24 gives the value stated in eqn 3.7.
- Estimate the ground-state energy of helium using the variational principle. (The details of this technique are not given in this book; see the section on further reading.)

(3.6) *Calculation of integrals for the 1s2p configuration*

- Draw a scale diagram of $R_{1s}^{Z=2}(r)$, $R_{2s}^{Z=1}(r)$ and $R_{2p}^{Z=1}(r)$. (See Table 2.2.)
- Calculate the direct integral in eqn 3.31 and show that it gives

$$J_{1s2p} = -\frac{e^2/4\pi\epsilon_0}{2a_0} \frac{13}{2 \times 5^5}.$$

Give the numerical value in eV (cf. that given in the text).

(3.7) *Expansion of $1/r_{12}$*

For $r_1 < r_2$ the binomial expansion of

$$\frac{1}{r_{12}} = (r_1^2 + r_2^2 - 2r_1r_2 \cos \theta_{12})^{-1/2}$$

is

$$\begin{aligned} \frac{1}{r_{12}} &= \frac{1}{r_2} \left\{ 1 - 2\frac{r_1}{r_2} \cos \theta_{12} + \left(\frac{r_1}{r_2}\right)^2 \right\}^{-1/2} \\ &\simeq \frac{1}{r_2} \left\{ 1 + \frac{r_1}{r_2} \cos \theta_{12} + \dots \right\}. \end{aligned} \quad (3.36)$$

(When $r_1 > r_2$ we must interchange r_1 and r_2 to obtain convergence.) The cosine of the angle between $\mathbf{r}_1$ and $\mathbf{r}_2$ is

$$\begin{aligned} \cos \theta_{12} &= \hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2 \\ &= \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 \cos(\phi_1 - \phi_2). \end{aligned}$$

- Show that the first two terms in the binomial expansion agree with the terms with $k = 0$ and 1 in eqn 3.30.
- The repulsion between a 1s- and an nl -electron is independent of m . Explain why, physically or mathematically.
- Show that eqn 3.32 leads to eqn 3.34 for $l = 1$.
- For a $1snl$ configuration, the quantity $K(r_1, r_2)$ in eqn 3.34 is proportional to r_1^l/r_2^{l+1} when $r_1 < r_2$. Explain this in terms of mathematical properties of the $Y_{l,m}$ functions.

Web site:

<http://www.physics.ox.ac.uk/users/foot>

This site has answers to some of the exercises, corrections and other supplementary information.